Pneumonia Identification on Chest X-ray Images Based on Deep Learning Models

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I. Introduction

One of the diseases that most people suffer from is pneumonia, a serious lung disease; nonetheless, it remains an ongoing concern for public health, notably impacting older people and young children [1]. The development of the treatment of patients depends on proper monitoring along with the correct screening; nevertheless, regular procedures for diagnosis based on radiologist manual analysis of chest X-ray images have major limitations. Commonly unreliable, at risk of mistakes by humans, and resource-intensive, these procedures could result in delays and fluctuation in diagnosis precision [2].

These difficulties have driven a boost in curiosity in controlled diagnostics using deep learning methods. In terms of their needs, Convolutional Neural Networks (CNNs) have evolved into an effective tool in healthcare image analysis as their improved recognition of features enables them to achieve impressive results in operations including image categorisation. About manual procedures, CNNs greatly reduce diagnosis delay and improve accuracy by enabling quick, reliable, and precise analyses [3].

The present research intends mainly to evaluate CNN architectures' diagnostic ability for chest X-ray image-based identification of paediatric pneumonia. The goal is particularly to generate a custom CNN model and evaluate its diagnostic effectiveness against four widely recognised pre-trained CNN models, including ResNet 18, DenseNet 121, EfficientNet-B0, and VGG16. Even though each of the five CNN models has been utilised, this analysis highlights, through the combination of quantitative metrics and qualitative evaluation, a comprehensive analysis of performance between the custom-built CNN and the two best-performing pre-trained models, which are DenseNet121 and VGG16 [4] [5] [6] [7].

The standardisation of greyscale paediatric chest X-ray pictures, which are resized to 224 by 224 pixels, describes the main mechanism of this algorithm. The CNN models then classify each image into one of two different categories, such as Normal or Pneumonia. Therefore, output recommends an ultimate binary diagnostic prediction, which is helpful and vital for quick clinical decision-making once the classification has been chosen.

To start with, a fundamental understanding is proposed by the substance of critical research, which is the foundation upon which this work expands. For instance, ResNet has been designed by He et al, which included residual connections. This design helped to improve the problem of disappearing gradients, which ultimately led to an advancement in the implementation of deep CNNs [4]. Additionally, Huang et al came up with DenseNet, which modified both generalisation and accuracy by demonstrating dense layer connections as an approach to enhancing the utilisation of features and reducing values for parameters [5]. Moreover, Tan and Le made use of compound scaling methodologies in order to build EfficientNet. Thus, all of these improvements are intended to optimise system performance as well as outcomes in the presence of many computational limits [6].

Likewise, the VGG16 structure design has been introduced by Simonyan and Zisserman, which proves the usefulness of deep yet basic CNN architectures in image classification applications [7]. Even though these designs have repeatedly proven high implementation across broad image datasets, reasonable studies, especially targeting paediatric pneumonia diagnosis, persistently remain inadequate. Overall, to overcome this knowledge inequality, this paper desperately explores various CNN structural designs within the unique setting of paediatric chest radiography, leading towards more applicable computerised pneumonia detection.

II. Dataset

The dataset employed in this study is the "Chest X-Ray Images (Pneumonia)" dataset, publicly available on Kaggle. It comprises 5,863 anterior-posterior (AP) chest X-ray images in JPEG format, obtained from paediatric patients at Guangzhou Women and Children's Medical Centre. The images are classified into two distinct categories: "Pneumonia" and "Normal." For the experiments conducted in this project, the dataset was partitioned into training, validation, and test subsets containing 5,216, 521, and 523 images, respectively. The distribution of samples is relatively balanced between the classes, facilitating effective training and unbiased evaluation.

In this experiment, the publicly available "Chest X-Ray Images (Pneumonia)" dataset has been employed, which can be found on Kaggle [8]. This dataset contains 5863 anterior-posterior (AP) chest X-ray JPEG-formatted pictures, all of which pictures collected from the paediatric patients at Guangzhou Women and Children's Medical Centre. Moreover, all the X-ray images were divided into two classes, including "Pneumonia" and "Normal" as can be seen in Figure 1. Plus, the dataset was split into training, validation, and test subsets, each of which had 4172, 521, and 523 photos, respectively, therefore enabling the implementation of the experiments conducted for this project because effective training and evaluation are made easier by the somewhat balanced sample distribution throughout the classes.

Additionally, a few necessary pre-processing procedures need to be done before starting to train all the models. These critical procedures include resizing all images, as each of the images is resized to the desired resolution of 224*224 pixels in order to meet the input requirements of all five CNN models. Moreover, the whole dataset was standardised utilising mean and standard deviations derived from ImageNet data, so normalising pixel intensities. Different forms of data enhancement on only the training set, including random cropping, horizontal flipping, and minor colour jitter adjustment, enhanced generalisation and prevented overfitting

throughout model development. Finally, these preprocessing steps greatly augmented the resilience of the trained CNN models, consequently developing their capacity to accurately classify pneumonia cases in undetectable chest X-ray images across test time.

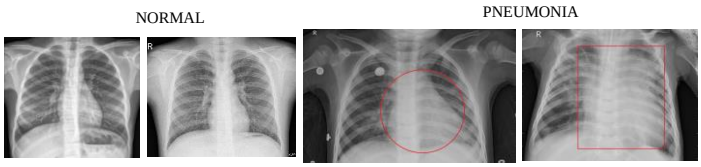


Figure 1 - Xray images

III. Methodology

A. Model

This research project generates a comparison between CNN and two pre-trained top-performing models; however, as mentioned earlier, the Python code is created with five models, but three of the models will be compared in this paper. The first model is an indicated CNN architecture including numerous convolutional layers with deepening filter layers in each other. Following the first model, which is CNN, ReLu activation, batch normalisation, dropout layers for binary classification and max pooling layers comprise the second model, as the architecture is shown in Figure 2. Then the third model features constructed by Batch Normalisation layers and Relu activation.

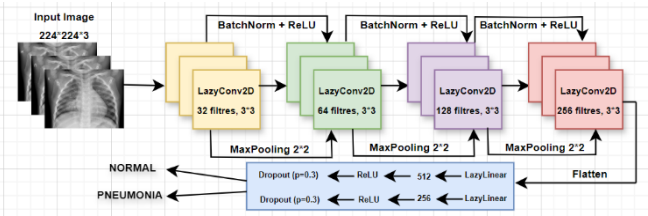


Figure 2 - custom CNN structure

Moreover, pre-trained on the ImageNet dataset, which is well-known for its effectiveness in picture classification tasks, was used for two pre-trained deep convolutional neural network (CNN) models, VGG16 and DenseNet121. However, unlike the closely connected layers of DenseNet121, which maximise features over several levels, VGG16 advantages a narrow yet deep stack of convolution layers, as the VGGQ6 architecture can be seen in Figure 3. Likewise, pre-trained and fine-tuned for the particular goal of pneumonia recognition using paediatric X-ray chest images. Pseudo codes for three models mentioned in the Appendices section which demonstrate the high-level summary of the implementation logic of all three models.

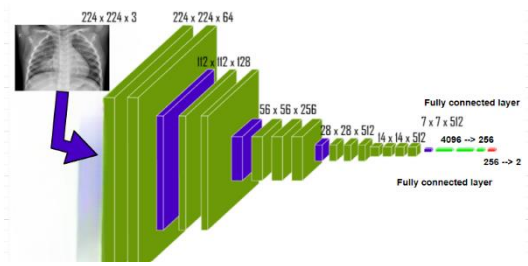


Figure 3 - VGG16 architecture

B. Loss

The Cross-Entropy Loss function has been used for all models in terms of training to enhance the learning procedure. The Cross-Entropy Loss function is shown below, and it is beneficial for classification tasks because it perfectly calculates the predicted probability distribution and the actual classes.

$$\boldsymbol{L} = -\sum_{j} \boldsymbol{Y_j} \log(\hat{\boldsymbol{Y}}\boldsymbol{j})$$

In the Cross-Entropy Loss equation above $\boldsymbol{Y_j}$ and $\hat{\boldsymbol{Y}}\boldsymbol{j}$ are the ground truth and the predicted probability, respectively, for class **j**. Likewise, using this formula helps the output of our model to get precise predictions rather than wrong predictions from the actual label. Also, in the code section, this function is implemented as `torch.nn.CrossEntropyLoss()` [9]. Therefore, the deployment of logarithmic terms guarantees that the consequence will exponentially expand if the confident forecasts are inaccurate, consequently pointing the model to acquire more trustworthy decision bounds.

C. Optimisation

In this section, the Adam optimiser has been used because it is very beneficial for the model to handle noisy data, and also, the other advantage of this optimiser is very adjustable to working with learning rates. To obtain the desired results output for all models, different learning rates have been trained and analysed for this experiment, where the lower learning rate of 0.0001 has been used custom CNN to allow for gradual convergence. However, the other two, including VGG16 and DenseNet121, have been trained with a learning rate of 0.001. Followed by choosing the optimiser, the early stopping is applied to the training section with the 32 images for the batch size, which has been done with 50 epochs for the custom CNN model and 10 epochs for VGG16 and DenseNet121 models, with a patience of 3 epochs to terminate the training process on a good accuracy and when the validation loss concluded to improve. Therefore, by all these added steps in the optimisation process, models won't overfit.

Overall, all these steps, including validation monitoring, supervised the learning rates with a desirable optimiser, improved the model's design and output result.

D. Discussions

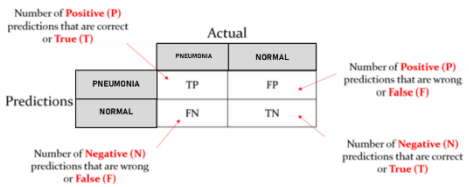
The comparison between the custom CNN and the two top-performing pre-trained models (VGG16 and DenseNet121) provides insight into accuracy, training complexity, and generalizability trade-offs [7] [5]. The pre-trained models benefit significantly from learned feature extraction, leading to rapid merging and advanced generalisation. The custom CNN, trained from scratch, offers transparency and straightforward interpretability but requires extensive training data and optimisation to achieve comparable performance. Despite these challenges, the Custom CNN provided competitive accuracy, suggesting that strategic architecture design and optimisation play crucial roles in its effectiveness.

IV. Experiment

A. Evaluation Metrics

The performance of all models was assessed utilising five necessary classification metrics, including accuracy, precision, recall, F1 score, and the area under the ROC curve (AUC). Subsequently, taking all of these parameters into consideration, a full evaluation of the models' proficiency in recognising pneumonia from chest X-ray pictures is provided.

Each of the classification metrics has a duty; for example, accuracy evaluates the complete acceptability or perfection of predictions by monitoring the ratio of positive recognitions, which are truly precise. And the advantage of recall is its power to express the quantity of models to recognise all possibilities of actual positive occurrences. Moreover, an acceptable evaluation will be approached with the F1 Score through the combination of recall and accuracy, which is exceptionally valuable when the dataset is imbalanced. Also, in this experiment, AUC-ROC has been applied to assess the effectiveness of models throughout classification thresholds, which indicates a balance between specificity and sensitivity. Classification metrics have been calculated through the equations below in:



$$ACCURACY = \frac{number\ of\ correct\ predictions}{total\ number\ of\ predictions} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$PRECISION = \frac{number\ of\ true\ positives}{number\ of\ predicted\ positives} = \frac{TP}{TP + FP}$$

$$RECALL = \frac{number\ of\ true\ positives}{number\ of\ actual\ positives} = \frac{TP}{TP + FN}$$

$$F1\ SCORE = 2 * \frac{(precision * recall)}{precision + recall}$$

B. *Quantitative Results*

The table below shows the summary of accuracy, precision, recall, F1 score, and AUC of all models obtained through training and evaluating on the Chest X-ray Images dataset.

As can be seen in the table below, overall VGG16 gained the most excellent functioning compared to other models by achieving an accuracy of 94.83%, precision, recall, and F1 score all at 0.9484, and an AUC of 0.99. Although the results of DenseNet121 are close to those of VGG16, by gaining an accuracy of 94.26%, precision, recall, and F1 score all at 0.9426, and an AUC of 0.99. However, the other two pre-trained models, including EfficientNet-B0 and ResNet18, obtained the same accuracy of 93.11%, where their AUCs are 0.98 and 0.97, respectively. Additionally, the custom CNN, which is built from scratch, acquired the accuracy of 91.97% and AUC of 0.98, which are a little bit lower than other pre-trained models but still competitive. Hence, the custom CNN will be compared to VGG16 and DenseNet121 in the following sections based on their reasonable results and performance.

Performance Metrics Of All Models					
Model	**Accuracy**	**Precision**	**Recall**	**F1 Score**	**AUC**
ResNet18	0.9311	0.9311	0.9311	0.9311	0.97
DenseNet121	0.9426	0.9426	0.9426	0.9426	0.99
EfficientNet	0.9311	0.9311	0.9311	0.9311	0.98
VGG16	0.9483	0.9484	0.9484	0.9484	0.99
Custom CNN	0.9197	0.9197	0.9197	0.9197	0.98

In terms of a decent understanding of how the classification works, the ROC curves of all three models are drawn from their obtained results as can be seen in Figure 4. The ROC curves show that among all thresholds, VGG16 and DenseNet121 own the competent classification performance, as their ROC curves are located in the upper left corner. Although its AUC was somewhat lower, the custom CNN also demonstrated exceptional ROC performance. Overall, the low false positive rates (FPR) and high true positive rates (TPR) are shown by these almost spotless ROC curves (AUC ~ 0.99).

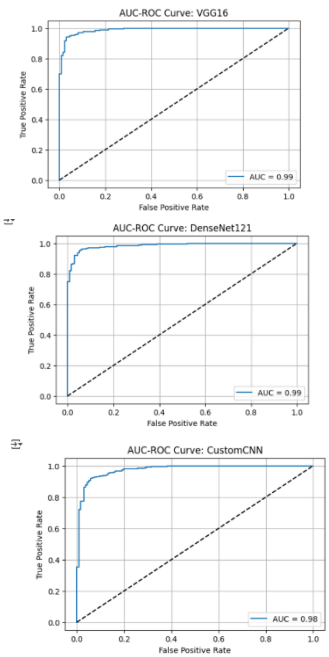


Figure 4 - AUC ROC results of three models

Following all the obtained quantitative results so far, the confusion matrix has been utilised for CNN, VGG16, and DenseNet121 models as shown in Figure 5. Based on the confusion matrix, VGG16 precisely predicted 125 normal and 371 pneumonia cases, with only 11 false positives and 16 false negatives, which is the highest correct prediction compared to the other two models; however, DenseNet121 results look the same as VGG16. On the other hand, the CNN model demonstrates somewhat greater misclassification, but still explainable and sensible.

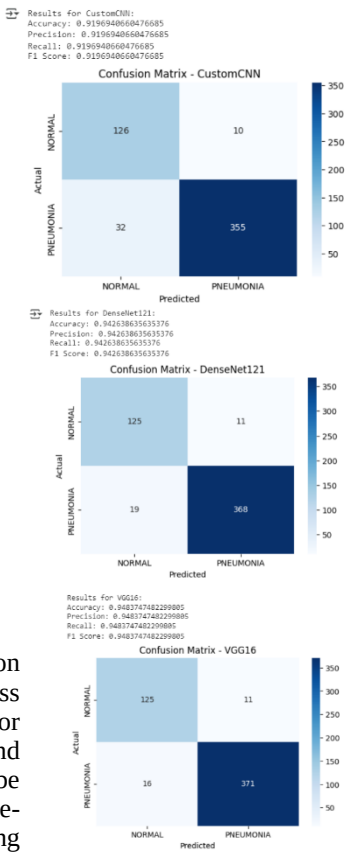


Figure 5 - confusion matrix results of three models

In the following section blow, the accuracy and loss curves has been examined for CNN, VGG16, and DenseNet121 models as can be seen in Figure 6. Both pre-trained models including VGG16 and DenseNet121 demonstrating quickly merging within 6 epochs which is outstanding, where the validation loss has been reduced to a expected level for both VGG16 and DenseNet121 models. On the other hand both models obtained the validation accuracy of 94.05% and 93.28%, respectively which they both are maintained at a reasonable level of accuracy and modest loss at epoch 6. Plus, CNN hit the validation accuracy of 91.75% which it proceeded steadily within epoch 9.

Additionally, batch normalisation and drop out which helped to improved generalisation and attempting to decrease overfitting, are reliable for this steady improvement.

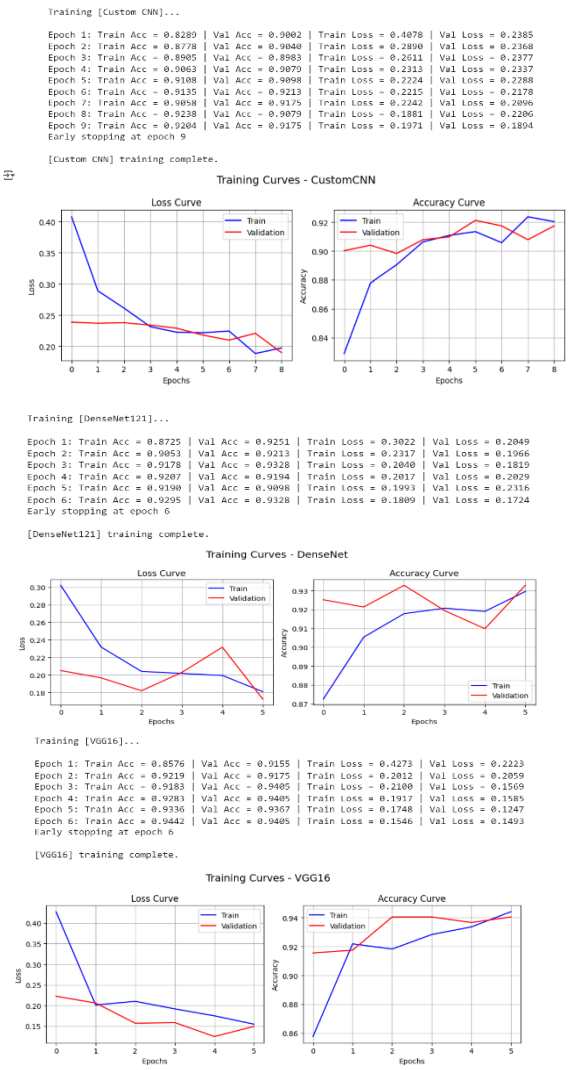


Figure 6 - validation loss and accuracy results

C. *Qualitative Results*

In terms of qualitative results, the Gradio web application was created to assess the pictures related to chest X-rays from the code, which was launched earlier, and to deliver a prediction. The Gradio web application performs predictions through five deep learning models, which were introduced in the previous section. So, all users could submit the chest x-ray picture and get the prediction as "Normal" or "Pneumonia", while users have the option to choose one of the five models, where all the results are represented in a bar chart with percentages [10].

A normal and a pneumonia picture have been assessed in the Gradio web application to prove the accuracy of models, as can be seen in Figure 7 and Figure 8. The normal picture has been evaluated with VGG16, and the model shows an acceptable prediction with 98% precision. Likewise, another picture for pneumonia has been assessed with the custom CNN model, and the model demonstrates an excellent result with a 100% accuracy, which is a fabulous prediction.

The examples of Gradio web applications represent the models' categorisation and real-time prediction performance in terms of accuracy, which means that all predictions correspond to ground truth labels and determine important generalisation on unseen test data.

D. *Comparisons*

In this paper comparison has been applied between two best-accomplishment pre-trained models, including VGG16 and DenseNet121 custom CNN, which is built from scratch [11] [12]. Comparison has been made with quantitative and qualitative results for each model.

- Quantitative Comparison

All the quantitative findings mentioned in the quantitative results sections clearly show that all pre-trained models approach a better performance with transfer learning when they are learning from a big-scale dataset. Though the custom CNN works competitively, specifically reflecting that the model was built and trained from scratch without any advantage of pre-trained weights.

- Qualitative Comparison

In contrast to quantitative results, qualitative evaluation is also vital because it provides clarification on the effectiveness of each model. For instance, in terms of model prediction accuracy, the Gradio web application has been built for the purpose of evaluating users' chest X-ray pictures.

For example, the VGG16 model properly predicted the normal picture of a chest x-ray with 100% accuracy, while the custom CNN model's accuracy of 98%, which was close to that of the VGG16 model in the example of the Gradio web application.

To conclude, in clinical circumstances, it is very critical to have high trust in predictions, especially when working with these kinds of deep learning models, and all the results prove that all the pre-trained models used in this research were reliable. Although the custom CNN also has a manageable and an outstanding performance in the race against pre-trained models, and demonstrates that it can be effective, especially in settings where processing resources are few.

- Validation through Training Dynamics

By comparing training and validation loss and accuracy, the model's performance can be evaluated and analysed. For example, the pre-trained models display how quickly they can merge in the first 10 epochs and then remain steady. But the custom CNN, compared to pre-trained models, was merged deliberately, and its best performance was in the 15th epoch.

Moreover, variants in training dynamics represent how transfer learning significantly speeds up the merging of models. But the custom CNN has been gradually becoming reliable, which means it can learn and generalise well, which proves its design is powerful.

Ultimately, both pre-trained models, including VGG16 and DenseNet121, represent exceptional implementation metrics, respectively, with an advantage of quickly merging. Additionally, the custom CNN shows that it can be competitive with pre-trained models, which can be useful in all cases of computational resources or approaches to large pre-trained models are restrained.

V. REFERENCES

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APPENDICES

```
Custom CNN:
Begin
  Load and preprocess dataset (resize, normalise)
  Split the dataset into training, validation, and test sets

  Define CNN model:
    - Convolutional Layer (filters=32, kernel size=3x3)
    - Batch Normalisation
    - ReLU Activation
    - Max Pooling
    - Repeat with increasing filters (64, 128, 256)
    - Flatten
    - Fully Connected Layer (512 units)
    - ReLU
    - Dropout
    - Fully Connected Layer (256 units)
    - ReLU
    - Dropout
    - Output Layer (2 units, Softmax)

  Compile the model with CrossEntropyLoss and Adam optimiser
  Train for 50 epochs with early stopping (patience = 3)
  Evaluate on test data
End
```

pseudo code for CNN

```
DenseNet121:
Begin
  Load and preprocess dataset (resize to 224x224, normalise)
  Split the dataset into training, validation, and test sets

  Load pre-trained DenseNet121 model
  Replace the classifier head with a custom output layer for 2 classes
  Optionally freeze early layers; fine-tune deeper layers

  Compile the model using CrossEntropyLoss and Adam optimiser
  Train for up to 50 epochs with early stopping (patience = 3)
  Evaluate model performance on the test dataset
End
```

pseudo code for DenseNet121

```
VGG16:
Begin
  Load and preprocess dataset (resize to 224x224, normalise)
  Split the dataset into training, validation, and test sets

  Load pre-trained VGG16 model
  Modify the final classifier layer to output 2 classes (pneumonia/normal)
  Freeze initial layers (optional fine-tuning of later layers)

  Compile the model using CrossEntropyLoss and Adam optimiser
  Train for up to 50 epochs with early stopping (patience = 3)
  Evaluate the model on test data
End
```

pseudo code for VGG16

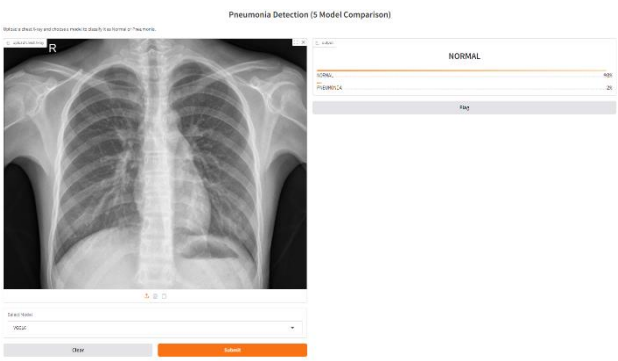

Figure 7 - normal xray image result

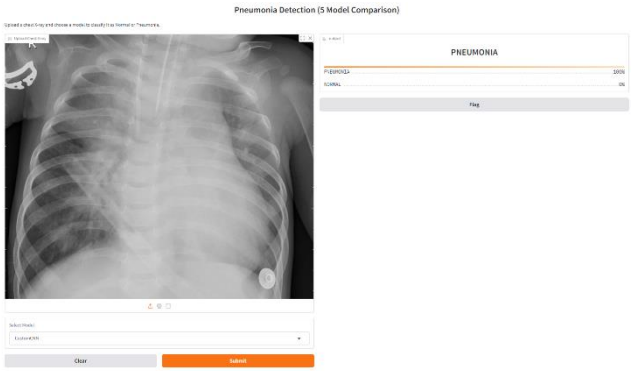

Figure 8 - pneumonia xray result